Reflectarray with Form-Birefringent Unit Cells

Deepdive on potential thesis topic

Online Version

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1 Reflectarray with Form-Birefringent Unit Cells

This document focuses on a potential topics of reflectarrays with unit cells that exhibit form-birefringerence – that is they exhibit anisotropic properties. It builds on the general reflectarray and material observations recorded in `reserach.typ` file.

2 The Idea: Form-Birefringent Unit Cells

2.1 Physical Mechanism

- Form birefringence creates an anisotropic medium using subwavelength features, so the incident wave does not resolve the individual physical boundaries and instead sees a homogenized anisotropic medium.
  - This can be achieved either with periodic cutouts or asymmetric voids in a single material, or by combining two or more isotropic materials into a subwavelength grating whose composite response is anisotropic [1].
  - Effective refractive index differs for electric fields polarized parallel ($n_{\parallel}$) versus perpendicular ($n_{\perp}$) to the structural axis.
- **Design rule for LP-to-CP conversion:**
  - When a linearly polarized wave is incident at $45°$ to the structural axis, its parallel and perpendicular components travel at different phase velocities and emerge with a differential phase delays.

$$\Delta\varphi = \frac{2\pi f d}{c_0}\left(n_{\parallel} - n_{\perp}\right)$$

  - Setting $\Delta\varphi = \frac{\pi}{2}$ (quarter-wave retardation) while keeping the two transmitted amplitudes roughly equal (This won't be possible to achieve.) converts the linear input into a circularly polarized output.

2.2 Reflectarray Application

- Reflectarray can be designed with unit cell that exhibits such anisotropy to have additional function aside from beam focusing
  - **Polarization transformation** (CP generation or LP-to-CP conversion).
  - **Polarization purification:** since most real feeders exhibit some cross-polar reception, the array can be designed so that only the desired polarization is focused, thereby improving cross-polar discrimination (XPD) of the overall system.
- Basic ideal for the new cell design comes from [2]
  - [2] achieved a flat reflectarray, comprised of unit cell with square footprint with variable square hole.
  - New cell design would have an elliptical (or rhombic) cutout in place of a symmetric void - with major axis of the body being oriented along the diagonal of the square.
  - the ellipse/rhombus presents a different effective permittivity along its major and minor axes, and thus a different reflection phase per polarization.
  - Conventional polarization-controlling reflectarrays rely on blocks of differing height to obtain the per-axis phase, like [3]
  - What hasn't really been tried is constructing an array of uniform height where the two degrees of freedom (necessary to tune $n_{\perp}$ and $n_{\parallel}$ independently) are achieved in geometry changes in x and y directions
  - Such design could enable easier addition of some covering to make the final reflectarray smooth at the surface futher enhancing it's mechanical properties
  - Or possibly in order to improve printability the unit cell could be inverted – instead of elliptical cutouts printing elliptical pillars.

2.3 Geometry

2.3.1 Array Size

Using basic gain formula

$$G = \eta\frac{4\pi A}{\lambda_0^2},$$

a gain of the reflect array can be estimated. Assuming aperture efficiency of 40 %, closed to what was achieved by [2] using $0.3\lambda_0$ size cells, [4] achieved 50 % with $0.4\lambda_0$ size. Table bellow shows rough gain estimation for a square aperture (Since the print bed is square there's no reason to consider any other geometry).

Aperture Size (λ)	**Gain** (dBi)	**Size 10 GHz** (mm)	**Size 15 GHz** (mm)	**Size 20 GHz** (mm)	**Size 25 GHz** (mm)	**Size 30 GHz** (mm)
6	22.5	180	120	90	72	60
8	25.07	240	160	120	96	80
10	27.01	300	200	150	120	100
12	28.60	360	240	180	144	120
14	29.94	420	280	210	168	140
16	31.10	480	320	240	192	160
18	32.12	540	360	270	216	180
20	33.03	600	400	300	240	200

2.3.2 Unit Cell Size

All text bellow is taken from [5]. The standard rule for reflectarrays is that the unit cell periodicity should be $\leq \frac{\lambda_0}{2}$ at the design frequency to avoid grating lobes. Dimension of $\frac{\lambda_0}{2}$ seems to be the standard in other literature.

Periodicity	**Advantages**	**Disadvantages**
$\frac{\lambda_0}{2}$	Standard; sufficient phase range (300°); good for large apertures	narrower bandwidth
$\frac{\lambda_0}{3}$	Better aperture sampling; wider bandwidth; lower element losses	Reduced phase range; steeper phase slope (more quantization error)
$\frac{\lambda_0}{4}$	Even wider bandwidth; further loss reduction; better for small-to-moderate arrays	Phase range severely restricted; would necessitate multi-layer design or high ε_r material

Especially if the thesis will be focused on creating reflectarray with additional capabilities like polarization transformation it's not worth to consider other cell designs then with $\frac{\lambda_0}{2}$ base dimension.

In terms the heigh, higher cell will mean A) larger losses in the dielectric and B) being more susceptible to incidence angle changes.

2.3.3 Conclusions for this topic.

Typical FDM printer print bed is in the range of 200 to 300 mm. For SLA most are between 120 to 150 mm. As the thesis will be mainly research focused a aperture size could be on the lower end - between 8 to 12 λ_0 giving size that should be printable on FDM even at 10 GHz and SLA till 25 GHz assuming 150 mm print bed.

As for unit cell size, bandwidth here isn't the primary consideration. Usable range of frequencies will be already limited plenty by the range in which polarization conversion will work as intended. Thus unit cell smaller than $\frac{\lambda_0}{2}$ doesn't really make that much sense.

2.4 Problems and Open Questions

- **Phase-range coupling is the central difficulty:**
  - The unit cell must supply $\Delta\varphi_x = 360°$ while maintaining $\Delta\varphi_y - \Delta\varphi_x \approx 90°$ across the entire phase range.
  - This effectively requires a single cell geometry that has to cover range of 0° to 450° while the orthogonal axis tracks it with a constant offset.

- This is a substantially harder problem than a single-polarization element.
- Block-based CP transformers side-step this by offsetting the y component in height relative to x.

- **The phase range of the available reference designs is insufficient:**
  - Assuming a smaller, compact aperture fed by a reasonable illuminator, a reduced maximum phase of roughly 300° might be tolerable – but even that demands coverage out to about 390° in the converted axis.
  - Existing dielectric reflectarray elements fall short of this.
    - [6] reached only 270° with $\varepsilon_r = 2.75$ and needed a custom composite at $\varepsilon_r = 4.4$ to cover desired range.
    - [4] required $\varepsilon_r = 7.5$ to reach roughly 300°.
    - The core reference [2] covered only about 200° (The exact value is not stated in the thesis, I'm judging from the phase curves they've listed), though this is quite extreme case.
  - The gap between required and available phase range is therefore the primary feasibility risk, and it pushes the design toward utilizing materials with the highest workable permittivity.
- **Manufacturing feasibility is in tension with the required geometry:**
  - **SLA:** has the resolution and build area to realize a compact reflectarray at higher frequencies, but high-permittivity resins are prohibitively expensive.
  - **FDM:** the filament availability problem is largely solved, but print resolution is the limiting factor – the unit cells are small and the anisotropic cutout geometry is not straightforward to print.
    - While precision of nozzle positioning on more advance printers is in the ranges of single, tens micrometers nozzle diameter is normally 0.4 mm.
    - While 0.2 mm nozzles are available and I think that Zetamix filaments should still be printable with them, even at 0.2 mm the resolutions is far behind what SLA can achieve.
    - The narrow gaps at the tips of an elliptical cutout may fall below the printer's reliable feature size, causing inconsistent extrusion or incomplete fill; a rhombic cutout fares somewhat better but is not a complete solution.
    - In addition the addition of material on walls – print being slightly bigger then designed requires consideration during design of the print model - which will slightly differ from the optimized model.
  - Mitigations:
    - Lower the design frequency so that resolution is less critical, at the cost of build-area problems.
    - Invert the design so that the anisotropic features are pillars rather than voids, avoiding thin walls that must be dimensionally accurate.
- **Design challenges:**
  - Because the two polarizations are coupled through a single geometry, the per-axis reflection phases cannot be tuned independently.
    - Incidence angle stability is up to questions. I would assume that ellipse would fair better then rhombus, but how well will the phase delay properties be kept across the array is hard to estimate.
  - Computer optimization is required to converge on a cell that meets both the total range and the 90° offset simultaneously.
  - I don't think this will be that big of a problem - S parameters for many different configuration of major - minor axis combos can be computed, surrogate model with Gaussian Processes or Polynomial Response created and configurations for each phase delay discovered on that.

3 Materials

- The unit cell is an all-dielectric, additively manufactured, so the material system sets both the achievable permittivity contrast (and therefore the attainable phase range) and the loss.
- **FDM with Zetamix** $\mathrm{TiO_2}$ (up to $\varepsilon_r = 10$) is the leading candidate: uniform filler distribution, a reproducible permittivity ladder, and no sedimentation issue.
- **SLA with a manually loaded resin** offers resolution and build-area headroom but carries the sedimentation risk and an unverified printability assumption.

3.1 The Baseline-Polymer Permittivity Ceiling

- Unloaded SLA and FDM polymers usually exhibit low relative permittivity, roughly ε_r between 2.0 and 3.0.
  - This ceiling limits the phase range obtainable from geometry alone, which is exactly the constraint identified above.
  - A polarization-converting cell needs independent control of two orthogonal reflection phases, so the phase-range limitation is even more restrictive than for a single-polarization reflector.
- Micro-voids from high feedstock viscosity and the natural result of FDM layering process depress the bulk dielectric constant.
- Hygroscopic polymers take up moisture, which further raises $\tan\,\delta$ and lowers ε_r.

3.2 High-Permittivity FDM Filaments

- High-permittivity filaments are made by blending the polymer matrix with ceramic nanoparticles such as barium titanate, titanium dioxide, or alumina.
  - For FDM systems the ceramic filler does not sediment as it does in liquid photopolymer systems, so the filler distribution remains uniform across the print.
- [7] offers a family of $\mathrm{TiO_2}$-loaded materials with ε_r of 2.2, 4.5, 7.5, and 10 at 100 % infill.
  - At 35 % infill the nominal $\varepsilon_r = 7.5$ material drops to around $\varepsilon_r = 2$.
- [6] obtained comparable material by manually adding submicron additives to a base PREPERM filament.
- Practical conclusion for this thesis:
  - The Zetamix $\mathrm{TiO_2}$ family is the most reproducible FDM option.
  - It's also easiest to get filament with high permittivity for any plausible 3D printer type.
  - The $\varepsilon_r = 10$ grade provides headroom to test whether the required phase shifts are even realizable before committing to a lower-permittivity material.

3.3 Photopolymer and Ceramic-Loaded Resins (SLA)

- In SLA and other liquid-resin processes, dense ceramic fillers undergo gravity-driven sedimentation before UV curing.
  - This induces an unintended anisotropic permittivity gradient along the vertical build axis.
  - Non-uniform particulate distribution also inhibits the maximum achievable effective permittivity.
- Most ceramic-loaded photocurable resins do not reach high permittivity because the additive concentration is kept small
  - [8] indicates values around $\varepsilon_r \approx 4$.
  - This is below what the phase-range analysis demands.
- Higher permittivity resins are created by loading base resin with $\mathrm{BaTiO_3}$ nanoparticles.
  - Manufacturer data for 3Dresyn HP1 BTO70 lists ε_r between 8 and 9 with loss tangent below 0.04 across 10-20 GHz range [9].
  - However, they are prohibitively expansive.
  - Loss factor of these resins is usually rather high – base resin is already fairly lossy, and barium titanate suffers from additional dipole relaxation losses on higher frequencies.

4 Characterization of the Unit Cell

4.1 Simulation of Unit cell

4.2 Measurement of the Unit Cell TODO

- I need to measure phase delay in two principal axis at 90 degree incident angle illuminated by a lineary polarized source.
  - Waveguide out of question – incidence angle changes with frequency.
  - Dielectric waveguide esque structure could maybe be used.
  - Free space to complicated.
  - Parallel plate?

5 Feasibility Study

- Aim's to verify if such unit cell can be realized with reasonable parameters using FDM printer with filament permittivities bellow $\varepsilon_r < 10$
  - The pillared variant is preferred because it improves printability: it avoids the thin, dimensionally-critical walls of an elliptical cutout.
- The study must answer two questions:
  - **A) Phase range on a single axis:** can the geometry produce a sufficient range of phase delays (target $\geq$ 300°) along one polarization?
  - **B) Cross-axis offset:** can a stable 90° phase difference be maintained between the two axes across that range?

5.1 Test Geometry

- An all-dielectric unit cell with:
  - A square footprint of $\frac{\lambda_0}{2} \times \frac{\lambda_0}{2}$ where $P = \frac{c_0}{2f_0}$.
  - An elliptical pillar whose major axis is aligned with the **diagonal** of the square, i.e. rotated 45° relative to the array axes.
  - A base substrate layer below the pillar.
  - $\varepsilon_r \in [7, 10]$; the headroom value used in the study is $\varepsilon_r = 9$.
- Because the ellipse presents a different projected cross-section to the x- and y-polarized waves, the two polarizations see different effective permittivities. This is the form-birefringence mechanism.

5.2 CST Setup

5.2.1 Create the project

- `File -> New -> Microwave & RF / Optical -> Unit Cell`.
- Set the frequency range (e.g. 25-35 GHz for Ka-band) and units (mm, GHz, ns).

5.2.2 Define parameters

- Open `Home -> Parameter List` (`Ctrl + P`) and add:

Name	Value	Description
`f0`	30	Design frequency (GHz)
`P`	5.0	Unit cell period, $\frac{\lambda_0}{2}$ at 30 GHz
`epsilon_r`	9	Material permittivity
`a`	1.5	Ellipse semi-major axis (mm)
`ratio`	2	Aspect ratio $\frac{a}{b}$, always > 1
`b`	`a / ratio`	Ellipse semi-minor axis (mm)
`H_pillar`	3.8	Pillar height (mm)
`H_base`	0.6	Base substrate thickness (mm)

- Defining `b = a / ratio` with `ratio > 1` enforces $a > b$ automatically, so the sweep only searches half of the parameter space.

5.2.3 Build the geometry

- **Ground plane:** `Modeling -> Brick` (`Ctrl + B`), name `ground`, X and Y from `-P/2` to `P/2`, Z from `-H_base` to `0`, material `PEC`.
- **Base dielectric:** `Ctrl + B`, name `base_dielectric`, X and Y from `-P/2` to `P/2`, Z from `0` to `H_base`, new material `Zetamix_eps9` with Epsilon set to `epsilon_r`.

- **Elliptical pillar:** CST has no native elliptical-cylinder primitive, so extrude a 2-D ellipse:
  - `Curves -> Ellipse`: centre `(0, 0, 0)`, radius X = `a`, radius Y = `b`.
  - `Curves -> Transform -> Rotate`: select the curve, axis z, angle `45`.
  - `Modeling -> Extrude Curves`: select the rotated ellipse, height `H_pillar`, name `pillar`.
  - `Modeling -> Transform -> Translate`: `pillar`, Z = `H_base`.
- **Boolean union:** `Modeling -> Boolean -> Add`, first `base_dielectric`, then `pillar`.

5.2.4 Boundaries and ports

- The Unit Cell template pre-configures these; verify them:
  - `Simulation -> Boundaries`: X and Y set to `unit cell`, Zmin set to `electric (Et = 0)`, Zmax set to `open (add space)`.
  - `Simulation -> Waveguide Ports`: the Floquet port exists. Open its properties and ensure Mode 1 (TE00) and Mode 2 (TM00) are enabled.
  - In the port's **Polarization** tab, set Mode 1 polarization angle to 0° and Mode 2 to 90°, so `S1,1` is the x-pol reflection and `S2,2` the y-pol reflection.
- **Caveat:** because the pillar sits at 45°, the structural eigenmodes are rotated relative to the global axes. Cross-terms `S1,2` and `S2,1` will therefore be non-zero. This is an expected consequence of the birefringence, not an error.

5.2.5 Solver

- `Simulation -> Frequency Domain Solver`.
- Accuracy `-40 dB` (coarse), frequency sampling set to **Fixed frequency list** at `30`, since single-frequency solves are much faster for sweeps.
- Mesh: adaptive refinement on, max passes `2`.
- Run and confirm `|S11|` is near 0 dB.

5.3 Sweep 1: Phase range on one axis (Question A)

- `Simulation -> Parameter Sweep`.
- Sweep `a` from `0.3` to `2.0` mm in steps of `0.3` (about 6 samples); keep `ratio` fixed (e.g. `2.5`).
- Result template: `S1,1` and `S2,2`, complex.
- Enable **Combine all sequences in one run**, then Start.
- Processing the results:
  - In the **Navigation Tree**, expand `1D Results -> S-Parameters`, right-click `S1,1` and choose **Plot Properties**.
  - Right-click the plot, choose **Export Data to File**, ASCII, **Export all curves in one file**, and save as `sweep_phase_x.txt`; repeat for `S2,2`.
  - Alternatively create result templates `phi_x = phase(S1,1)` and `phi_y = phase(S2,2)` via `Post-Processing -> Result Templates -> New -> 1D Result from 1D Results`, then export those directly in degrees.
- Build the assessment table:

a (mm)	φ_x (°)	φ_y (°)	$\Delta\varphi = \varphi_y - \varphi_x$ (°)
0.3	..	..	..
0.6	..	..	..
..	..	..	..

- Go/no-go metrics:
  - φ_x range, $\max(\varphi_x) - \min(\varphi_x)$, target $\geq 300°$.

  - φ_y range, target $\geq 390°$, i.e. $300° + 90°$.
  - $\Delta\varphi$ stability, target within $90° \pm 15°$ over most of the sweep.
- **Note:** CST returns phases in $(-180°, 180°]$. Always unwrap before computing ranges, otherwise a wrap past $\pm 180°$ will corrupt the result.

5.4 Sweep 2: Cross-axis offset (Question B)

- `Simulation -> Parameter Sweep` with two parameters:
  - `a` from `0.5` to `2.0` mm in steps of `0.3`.
  - `ratio` from `1.5` to `4.0` in steps of `0.5`.
- This is about 36 simulations. Use **Sequential** mode so all combinations are covered.
- Because `b = a / ratio`, the constraint $a > b$ is preserved with no extra work.
- Processing:
  - Create 2-D scalar templates via `Post-Processing -> Result Templates -> New -> 2D Scalar Result from 1D Results`:
    - `phase_range_x`: `max(phase(S1,1)) - min(phase(S1,1))`, parameters `a` and `ratio`.
    - `phase_range_y`: `max(phase(S2,2)) - min(phase(S2,2))`.
    - `phase_diff_mean`: `mean(phase(S2,2) - phase(S1,1))`.
  - Plot each from `Results -> 2D Results -> ...` as a heatmap: bright regions where the phase range exceeds the targets, and a region near 90° for `phase_diff_mean`.
  - The overlap of the feasible regions in the range map and the 90° map defines the usable $(a, :)$ design space.
  - Right-click a 2-D plot, choose **Export -> Export to File** (ASCII matrix) for the design map.

5.5 Unwrap check (external script)

- After exporting the raw phase curves, verify the numbers outside CST:

```
import numpy as np
data_x = np.loadtxt('sweep_phase_x.txt', skiprows=1)
data_y = np.loadtxt('sweep_phase_y.txt', skiprows=1)
phases_x = np.unwrap(data_x[:, 1:], axis=0)
phases_y = np.unwrap(data_y[:, 1:], axis=0)
for i in range(phases_x.shape[1]):
    phi_x, phi_y = phases_x[:, i], phases_y[:, i]
    delta = phi_y - phi_x
    print(f"cfg {i}: phi_x range={np.ptp(phi_x):.1f}, "
          f"phi_y range={np.ptp(phi_y):.1f}, "
          f"delta mean={np.mean(delta):.1f} +/- {np.std(delta):.1f}")
```

5.6 Interpretation and next steps

- **If all three criteria pass:** the concept is feasible. Proceed to a finer sweep around the best $(a, :)$ region and then to the surrogate-model optimization described in the design-challenges note (Gaussian Process / polynomial response surface).
- **If the phase range falls short:** the levers are a taller pillar, a higher ε_r (up to 10), and a smaller-than-$\frac{\lambda_0}{2}$ footprint, which improves range at the cost of grating-lobe margin ([5]).
- **If the 90° offset drifts:** the diagonal orientation alone may couple the axes. Consider adding a second geometric degree of freedom, as [3] added independent heights, to decouple the per-axis phases.

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